



Semiconductor chip containing hundreds of electron superhighways

tor that's commonly used in making night-vision goggles is called 'quantum spin Hall topological insulator.' It is one of the building blocks needed to create quantum particles that store and manipulate data. Topological designs are expected to be more fault-tolerant than other types of quantum computers because each qubit in a topological quantum computer will be made from a pair of quantum particles that have a virtually immutable shared identity. The catch to the topological approach is that physicists are yet to create or observe one of these stable pairs of particles, which are called 'Majorana fermions.'

Topological insulators are oddities; although electricity cannot flow through these, it can flow around their narrow outer edges. If a small square of a topological insulator is attached to a superconductor, elusive Majorana fermions are expected to appear precisely where the materials meet. The devices could potentially be used to generate qubits for quantum computing.

Nanostructure technology to make thin-film solar cells

Researchers have exploited an advanced nanostructure technology to make efficient and cheaper silicon solar cells. Developed jointly by Nanyang Technological University (NTU) and A*STAR Institute of Microelectronics (IME), the nano-structured solar cells can produce a current of 34.3 mA/cm². The resulting electricity current output is close to that of traditional cells (40 mA/cm²).

Conventional thin-film solar cells usually produce about half the current produced by traditional cells. The newly-developed nanostructure method creates a unique texture on the surface of amorphous silicon, improving the power-conversion efficiency of the thin-film silicon cell and hence the energy output.

"To mitigate against reduced light absorption and carrier recombination in the amorphous silicon thin-film cells, we designed and fabricated the novel nanostructures on silicon surface. The sole application of the surface texturing strategy achieved a record high of short-circuit current density with 5.26 per cent power-conversion efficiency," said Dr Navab Singh, senior scientist of IME's NanoElectronics Programme and leader of the project.

Antennae in your clothes?

The next generation of communication devices could be more reliable, thanks to Ohio State University researchers

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